

1. Generate $N = 500$, 2-D random data points and plot its corresponding Gaussian PDF.

$$f(x, y) = \frac{\exp\left\{\frac{-1}{2(1-\rho^2)}\left[\left(\frac{(x-\mu_x)}{\sigma_x}\right)^2 - 2\rho\left(\frac{(x-\mu_x)}{\sigma_x}\right)\left(\frac{(y-\mu_y)}{\sigma_y}\right) + \left(\frac{(y-\mu_y)}{\sigma_y}\right)^2\right]\right\}}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \quad (1)$$

$$T = \rho\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right) + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 \quad (2)$$

$$\mu = [0, 0]^T$$

- covariance matrix: $\begin{bmatrix} 1 & -0.8 \\ -0.8 & 1 \end{bmatrix}$
- covariance matrix: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- covariance matrix: $\begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$

2. Generate $N = 500$, 2-dimensional data points that are distributed according to the Gaussian distribution $N(m, S)$, with mean $m = [0, 0]^T$ and covariance matrix

$$S_1 = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$$

for the following cases:

- $\sigma_1^2 = \sigma_2^2 = 1, \sigma_{12} = 0$
 - $\sigma_1^2 = \sigma_2^2 = 0.2, \sigma_{12} = 0$
 - $\sigma_1^2 = \sigma_2^2 = 2, \sigma_{12} = 0$
 - $\sigma_1^2 = 0.2, \sigma_2^2 = 2, \sigma_{12} = 0$
 - $\sigma_1^2 = 2, \sigma_2^2 = 0.2, \sigma_{12} = 0$
 - $\sigma_1^2 = \sigma_2^2 = 1, \sigma_{12} = 0.5$
 - $\sigma_1^2 = 0.3, \sigma_2^2 = 2, \sigma_{12} = 0.5$
 - $\sigma_1^2 = 0.3, \sigma_2^2 = 2, \sigma_{12} = -0.5$
3. Considering iris dataset (Click here to download), design a Bayesian classifier to classify the test data. Divide the dataset into training and testing data into (a) 70:30 ratio (b) 80:20 ratio. Calculate the accuracy and error rate.